



Soutenance de thèse

Conception d'un système mémoire pour traitement de données creuses

Thèse soutenue par

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ÉCOLE DOCTORALE
EEATS, Spécialité NENT

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Outline

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Prérequis

- Début de ASAP ? Jusqu'à SpDC ?
- OP SpMV seulement ?

Cache de reduction

Utiliser la théorie pour montrer pourquoi on veut exploiter les structures des matrices

- Matrices creuses, éléments isolés -> cache devient fully associatif! Mais pas possible, et structure exploitable, de la matrice mais aussi kernel
- montrer dense spmv dans cache (introduire notion de thrashing) VS bande horizontale dans cache
- A quoi ressemblent les matrices ?
- Les matrices bandées
- On prend une bande, on la met dans un cache, si on choisit correctement la taille c'est ok -> DRC

SpDCache, fonctionnement + détails RTL

Evaluations (tout, python + RTL + théorie) Attention comp entre les 3 bancaire

Mettre l'accent sur l'approche plutôt que les résultats ?

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There Is No Largest Prime Number

The proof uses *reductio ad absurdum*.

Theorem

There is no largest prime number.

Démonstration.

- ❶ Suppose p were the largest prime number.
- ❷ But $q + 1$ is greater than 1, thus divisible by some prime number not in the first p numbers. □

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Thanks !